Age at Intake and Length of Stay Among Adopted Dogs at San Jose Animal Shelter (2018–2025)*

Not Just About Age: Rethinking Adoption Timelines

Anna Wadlow

October 8, 2025

San Jose Animal Care and Services (ACS) has faced sustained overpopulation, making it important to understand drivers of animals' length of stay (LOS). Using ACS open data from July 2018 to present, we examined whether a dog's age at intake predicts LOS among dogs whose outcome was adoption (n = 8,846). We fit a simple linear regression of LOS, in days, on age, in years. The estimated slope is small: 0.85 days per additional year, with an estimated intercept of 18.38 days, indicating only a weak, positive association. Model diagnostics show violations of linear model assumptions and a clear shift in residual patterns beginning around 2022. These patterns suggest unmodeled factors, such as intake condition, behavior, breed/size, or time-varying shelter operations, play a larger role than age alone. We conclude that age at intake, by itself, is a poor predictor of LOS at ACS and recommend incorporating additional variable into the model to better inform operational decision-making and welfare outcomes.

1 Introduction

San Jose Animal Care and Services (ACS) is a full-service shelter that has served San Jose and surrounding cities since 2004. In recent years, ACS has frequently operated over capacity, triggering an official city audit in the fall of 2024 (City Auditor 2024). Shelter population directly affects animal welfare and operational efficiency. Longer lengths of stay (LOS) increase the risk of disease — especially contagious respiratory infections — and constrain staff resources,

*Project repository available at: https://github.com/anna-wadlow/MATH261A-Project-1.

which can in turn lower adoption rates and contribute to the overpopulation (Maddie's Shelter Medicine Program 2018). At the extreme, prolonged LOS can intersect with population management tools, such as euthanasia.

What this paper aims to understand is if a dog's age at intake is associated with its LOS, among dogs ultimately adopted from ACS.

Prior work in shelter contexts generally treats age as one of several potential correlates of LOS, alongside factors such as intake type/condition, breed/size, color/appearance, behavior notes, and risk of euthanasia (Garrison and Weiss 2014). However, the effects of age can vary by facility and period, motivating an ACS-specific analysis that uses the shelter's own records to quantify the association. Given the overpopulation crisis of the SJ shelter, understanding the impacts of age, among other factors, to a dog's length of stay can inform remedial strategy.

We use the City of San Jose's open ACS data (updated daily) and restrict the analysis to dogs whose outcome was adoption between July 2018 and the present. A simple linear regression is used to relate LOS to age. We found a modest, positive estimated slope of about 0.85 additional days of LOS per year of age increase. But, diagnostic plots indicate the linear model performs poorly (heteroskedasticity, non-normal residuals, and time-related shifts) suggesting the need for a more robust model.

Section 2 describes the ACS data and key variables used. Section 3 outlines the linear modeling approach and Section 4 presents estimates and diagnostics. Section 5 interprets findings, reviews limitations, and proposes methodological and operational next steps.

2 Data

Data for this paper was obtained from the City of San Jose Open Data Portal ("Animal Shelter Intake and Outcomes - San Jose CA Open Data Portal," n.d.). ACS provides intake and outcome records for animals housed at their facility with daily updates covering July 1, 2018 through present. This paper focuses on dogs with an adoption outcome, yielding 8,846 records after data filtering.

The key variables, organized in columns, are animal type, date of birth (DOB), intake date, outcome date, and outcome type. Additional descriptors include the animals' primary breed, sex/sterilization status, primary/secondary colors, intake/outcome condition, intake type/subtype, and intake reason.

This paper looks at the records of those dogs who were adopted, which is one of the top three occurring outcomes in this dataset.

The proposed predictor variable, dog's age in years at intake is calculated from intake date and DOB variables. While ACS does not provide details of how DOB is obtained, it is reasonable to presume an estimate is often used. When the dog's DOB is unknown, commonly their age is estimated using various factors, such as ocular light reflection and dentition (Gesierich, Failing, and Neiger 2015), but these methods are not exact. In this dataset, the majority of ages are concentrated around whole-year values, suggesting that the recorded ages were likely estimated in full years rather than exact dates.

For the purpose of this paper, we use the DOB reported by ACS as a reasonable estimate. After data cleaning, the median age is 1.56 years, with minimum age 0 years and maximum age 19 years.

The model's dependent variable, length of stay, is the difference in days between the outcome date and intake date. After data cleaning, the median length of stay is 10 days, with minimum LOS 0 days and maximum LOS 214 days.

When plotted, Figure 1 shows quite a bit of variability between the dogs' age at intake and their length of stay at the shelter, somewhat similar to a graph of exponential decay. This prompts the question of whether a linear regression model is appropriate to address the research question of this paper.

It should be noted that a small percentage (0.016) of adopted dogs lacked data for date of birth and were excluded from the study. It is also unclear in the raw data if there are records for animals returned to the shelter after adoption, creating a "new" record. The 2024 city audit (City Auditor 2024) flagged opportunities for ACS to improve their data management and reporting, confirming the need for careful consideration when interpreting results.

The standard linear regression assumptions (independent errors, constant variance, normality of residuals, and linear relationship) are evaluated in Section 4.

3 Methods

We use a simple linear regression model, comparing length of stay (LOS) to age at intake.

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

The predictor variable, age of dogs at intake, is denoted by X_i and the response variable, length of stay, is denoted by Y_i.

The slope, β_1, represents the estimated change in length of stay per additional year of age. The intercept, β_0, represents the estimated length of stay for a zero-year-old dog. The residuals, ε_i, are assumed to be approximately normally distributed.

Using a simple linear regression model for this analysis may be overly rudimentary, ignoring other variables that can influence length of stay, but provides a baseline investigation.

We use the base R's `lm()` function to implement the linear model and the `confint()` function to calculate confidence intervals (R Core Team 2025) These are used alongside functions from

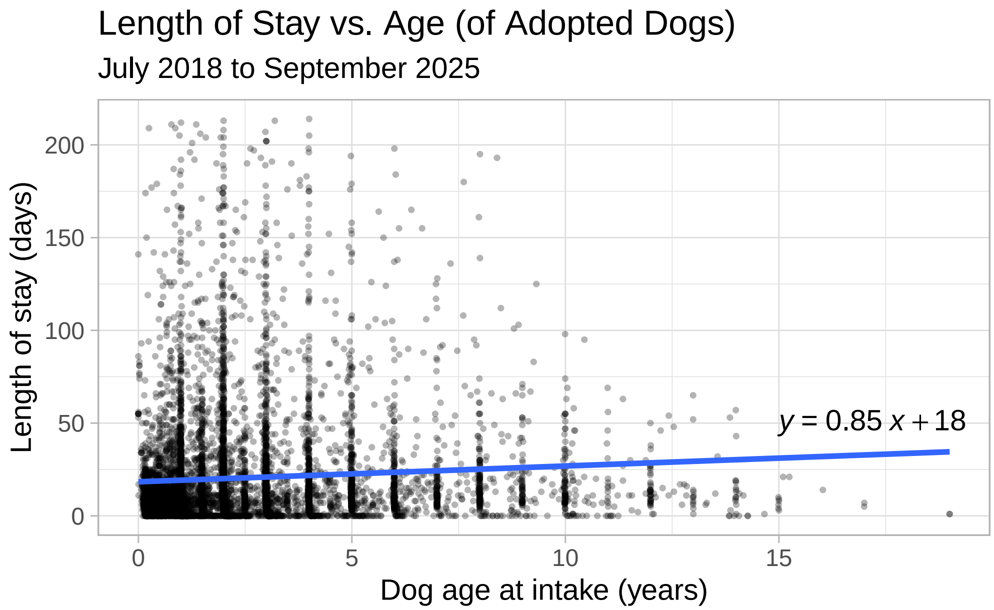

Figure 1: Scatterplot of dog age at intake versus their length of stay for adopted dogs. The fitted linear model, y = .85x + 18) indicates a slight positive association.

the `dplyr` (Wickham et al. 2023) and `tidyverse` (Wickham et al. 2019) packages for data cleaning and visualization.

Model fit was evaluated using diagnostic plots discussed in Section 4.

4 Results

The estimated slope parameter is $b_1 = 0.85$, which can be interpreted to mean that a one-year increase in dog's age, on average, corresponds to ~0.85 additional days in LOS. A 95% confidence interval for the estimated slope parameter b_1 is about [0.603, 1.098]. We estimate that 95% of the time, the true population slope is contained within this range. The estimated intercept is $b_0 = 18.38$, the expected LOS for a 0-year-old dog.

The results of the linear regression model should be approached cautiously since the residual plots Figure 2 show important assumption violations.

The residual vs. predictor plot shows that mean residuals appear positive at younger ages and trend negative at older ages, suggesting under-prediction for young dogs and over-prediction for older dogs. The variance of residuals is much larger among younger dogs, violating constant variance (heteroskedasticity). The clear non-linear shape of residuals also indicates a lack of linearity between our predictor and response variables.

Residuals plotted against intake date show a marked structural change beginning around 2022, consistent with time-varying shelter conditions and, possibly, data management shifts not captured by the model. This plot would suggest a lack of independence in the model errors.

Finally, the QQ plot Figure 3 shows pronounced right-skew and heavy tails, indicating residual non-normality. A potential cause for this is that both our predictor and response variables are bounded at zero.

While the p-value for the b_1 estimator is small ($1.7835594 \times 10^{-11}$), the R^2 value, 0.0051, combined with the violations of necessary assumptions cast serious doubt on the validity of this model and any associated tests.

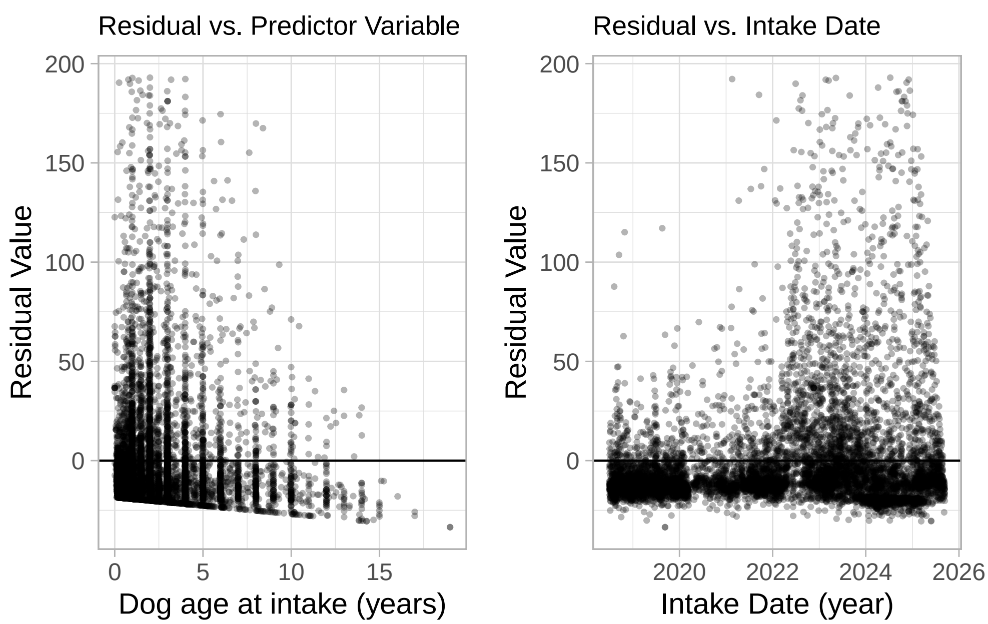

Figure 2: Residual diagnostic plots for linear regression model predicting a dog's length of stay based on their age at intake.

5 Discussion

Age at intake has, at best, a weak positive association with LOS among adopted dogs at ACS. Visualizations in Section 4 reinforce that LOS varies widely at all ages and that a straight-line

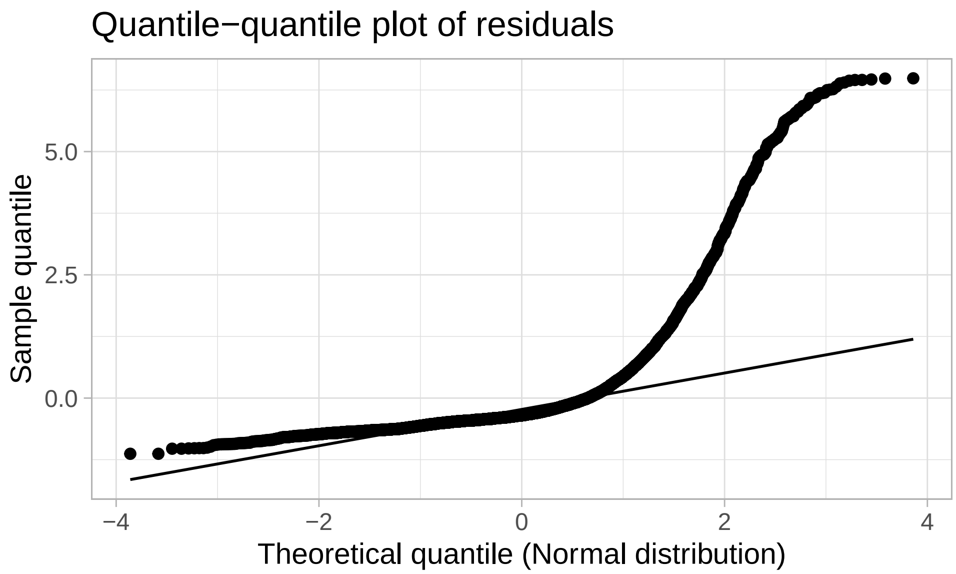

Figure 3: Diagnostic quantile-quantile plot indicates deviation from normality.

relationship is an oversimplification.

Consistent with previous research in this area, age at intake is a weak and unreliable predictor of LOS among adopted dogs at ACS. For operations and welfare decisions, relying solely on age as the predictor of length of stay would likely misdirect resources. A multi-factor profile should be explored to better understand the true relationship.

Several data limitations were identified and pose accuracy concerns of the model. Many dogs lack a recorded or reliable DOB and as such, these cases were removed, potentially introducing a selection bias. It is unclear whether each record reflects a unique dog or whether returns/re-adoptions appear as separate rows. More generally, the 2024 audit flagged opportunities to improve ACS data management and reporting.

As a modeling extension, additional variables should be considered to develop a better fitting model. Suggestions for these include the available descriptive variables: intake type/condition, sex/sterilization, breed/size, color/appearance, behavior notes, outcome subtype, and calendar time (e.g., year, month, pandemic/post-pandemic indicators).

Only dogs who were adopted were considered in the analysis for this paper. Given the underlying problem of overpopulation at San Jose's shelter, an extension of the research to assess "live" outcomes rather than adoption only, could be fruitful. This would expand the pool to include dogs who were returned to their owners and those released to an animal rescue organization.

In conjunction with the consideration of multiple predictors, shifting the model's goal to look at the probability of exceeding operational thresholds (e.g., LOS > 30 days) would better support capacity planning at ACS.

References

"Animal Shelter Intake and Outcomes - San Jose CA Open Data Portal." n.d. https://data.sanjoseca.gov/dataset/animal-shelter-intake-and-outcomes.

City Auditor, Office of the. 2024. "24.06 Animal Care and Services: Addressing Capacity Constraints and Updating Shelter Protocols Will Help Improve Animal Outcomes." https://www.sanjoseca.gov/home/showdocument?id=116106.

Garrison, Laurie, and Emily Weiss. 2014. "What Do People Want? Factors People Consider When Acquiring Dogs, the Complexity of the Choices They Make, and Implications for Nonhuman Animal Relocation Programs." *Journal of Applied Animal Welfare Science : JAAWS* 18 (August): 1–17. https://doi.org/10.1080/10888705.2014.943836.

Gesierich, K, Klaus Failing, and Reto Neiger. 2015. "Age Determination in Dogs Using Ocular Light Reflection, Dental Abrasion and Tartar." *Tierärztliche Praxis K: Kleintiere/Heimtiere* 43: 317–22. https://api.semanticscholar.org/CorpusID:7860692.

Maddie's Shelter Medicine Program, College of Veterinary Medicine. 2018. "Canine Respiratory Infections in Animal Shelters." *University of Florida*. https://sheltermedicine.vetmed.ufl.edu/files/2017/01/Canine-respiratory-infections-in-shelters.2018.pdf.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.

Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2023. *Dplyr: A Grammar of Data Manipulation*. https://doi.org/10.32614/CRAN.package.dplyr.